MongoDB Practical 1–25

Complete Questions, Separate Commands and Outputs

Easy practical record for mongosh / MongoDB

PRACTICAL 1

Question: Show all databases in MongoDB.

Command 1

```
show dbs
```

Output

```
admin
config
local
student
```

PRACTICAL 2

Question: Select the student database and show all collections.

Command 1

```
use student
```

Output

```
switched to db student
```

Command 2

```
show collections
```

Output

```
Faculty
Student
students
```

PRACTICAL 3

Question: Insert one student document into the Student collection.

Command 1

```
db.Student.insertOne({
  name: "Tulika",
  course: "BCA"
})
```

Output

```
{ acknowledged: true, insertedId: ObjectId('...') }
```

PRACTICAL 4

Question: Insert multiple student documents into the Student collection.

Command 1

```
db.Student.insertMany([
  {
    RollNo: 102,
    Name: "Tulika",
    Department: "BCA",
    Semester: 5,
    CGPA: 8.2
  },
  {
    RollNo: 103,
    Name: "Harsh Sharma",
    Department: "BCA",
    Semester: 5,
    CGPA: 9.1
  },
  {
    RollNo: 104,
    Name: "Aman Singh",
    Department: "BTech",
    Semester: 5,
    CGPA: 7.8
  },
  {
    RollNo: 105,
    Name: "Ritu",
    Department: "BCA",
    Semester: 4,
    CGPA: 8.7
  }
])
```

Output

```
{ acknowledged: true, insertedIds: { '0': ObjectId('...'), '1': ObjectId('...'), '2': ObjectId('...'), '3': Ob
```

PRACTICAL 5

Question: Display all student documents.

Command 1

```
db.Student.find().pretty()
```

Output

```
Displays all documents in the Student collection.
```

PRACTICAL 6

Question: Find a student using Roll Number.

Command 1

```
db.Student.findOne({ RollNo: 103 })
```

Output

```
{ RollNo: 103, Name: 'Tulika', Department: 'BCA', Semester: 5, CGPA: 9.1 }
```

PRACTICAL 7

Question: Find all students belonging to the BCA department.

Command 1

```
db.Student.find({
  Department: "BCA"
}).pretty()
```

Output

```
Displays all matching BCA student documents.
```

PRACTICAL 8

Question: Display only the Name and Department of BCA students.

Command 1

```
db.Student.find(
  { Department: "BCA" },
  { _id: 0, Name: 1, Department: 1 }
)
```

Output

```
{ Name: 'Tulika', Department: 'BCA' }
{ Name: 'Harsh Sharma', Department: '
BCA' }{ Name: 'SShikha', Department: '
BCA' }
```

PRACTICAL 9

Question: Update the department of Roll No. 102.

Command 1

```
db.Student.updateOne(
  { RollNo: 102 },
  { $set: { Department: "BCA-AI" } }
)
```

Output

```
{ acknowledged: true, matchedCount: 1, modifiedCount: 1 }
```

Command 2

```
db.Student.findOne({ RollNo: 102 })
```

Output

```
Department: 'BCA-AI'
```

PRACTICAL 10

Question: Update the semester of all BCA students to 6.

Command 1

```
db.Student.updateMany(
  { Department: "BCA" },
  { $set: { Semester: 6 } }
)
```

Output

```
{ acknowledged: true, matchedCount: 2, modifiedCount: 2 }
```

PRACTICAL 11

Question: Replace the complete document of Roll No. 103.

Command 1

```
db.Student.replaceOne(
  { RollNo: 103 },
  {
    RollNo: 103,
    Name: "Tulika",
    Department: "BCA",
    Semester: 6,
    CGPA: 9.3
  }
)
```

Output

```
{ acknowledged: true, matchedCount: 1, modifiedCount: 1 }
```

Command 2

```
db.Student.findOne({ RollNo: 103 })
```

Output

```
CGPA: 9.3
```

PRACTICAL 12

Question: Add an Email field to all students.

Command 1

```
db.Student.updateMany(
  {},
  { $set: { Email: "student@example.com" } }
)
```

Output

```
{ acknowledged: true, matchedCount: 5, modifiedCount: 5 }
```

PRACTICAL 13

Question: Remove the Email field from all students.

Command 1

```
db.Student.updateMany(
  {},
  { $unset: { Email: "" } }
)
```

Output

```
{ acknowledged: true, matchedCount: 5, modifiedCount: 5 }
```

PRACTICAL 14

Question: Delete the student with Roll No. 105.

Command 1

```
db.Student.deleteOne({ RollNo: 105 })
```

Output

```
{ acknowledged: true, deletedCount: 1 }
```

Command 2

```
db.Student.findOne({ RollNo: 105 })
```

Output

```
null
```

PRACTICAL 15

Question: Delete all students having CGPA less than 8.

Command 1

```
db.Student.deleteMany({ CGPA: { $lt: 8 } })
```

Output

```
{ acknowledged: true, deletedCount: 1 }
```

Command 2

```
db.Student.find({ CGPA: { $lt: 8 } })
```

Output

```
[]
```

PRACTICAL 16

Question: Count the total number of student documents.

Command 1

```
db.Student.countDocuments()
```

Output

```
3
```

Command 2

```
db.Student.find().pretty()
```

Output

```
Displays the 3 remaining student documents.
```

PRACTICAL 17

Question: Rename the Student collection to StudentDetails.

Command 1

```
db.Student.renameCollection("StudentDetails")
```

Output

```
{ ok: 1 }
```

Command 2

```
show collections
```

Output

```
Courses
Faculty
StudentDetails
students
```

PRACTICAL 18

Question: Create a Departments collection.

Command 1

```
db.createCollection("Departments")
```

Output

```
{ ok: 1 }
```

Command 2

```
show collections
```

Output

```
Courses
Departments
Faculty
StudentDetails
students
```

PRACTICAL 19

Question: Drop the Departments collection.

Command 1

```
db.Departments.drop()
```

Output

```
true
```

Command 2

```
show collections
```

Output

```
Courses
Faculty
StudentDetails
students
```

PRACTICAL 20

Question: Drop the complete student database.

Command 1

```
use student
```

Output

```
already on db student
```

Command 2

```
db.dropDatabase()
```

Output

```
{ ok: 1, dropped: 'student' }
```

Command 3

```
show dbs
```

Output

```
admin
config
local
```

PRACTICAL 21

Question: Create the student database and Students collection again, then insert three students.

Command 1

```
use student
```

Output

```
switched to db student
```

Command 2

```
db.createCollection("Students")
```

Output

```
{ ok: 1 }
```

Command 3

```
db.Students.insertMany([
  {
    RollNo: 201,
    Name: "Tulika",
    Department: "BCA",
    Semester: 5,
    CGPA: 8.5
  },
  {
    RollNo: 202,
    Name: "Shikha",
    Department: "BCA",
    Semester: 5,
    CGPA: 8.2
  },
  {
    RollNo: 203,
    Name: "Harsh Sharma",
    Department: "BCA",
    Semester: 5,
    CGPA: 9.0
  }
])
```

Output

```
{ acknowledged: true, insertedIds: { '0': ObjectId('...'), '1': ObjectId('...'), '2': ObjectId('...') } }
```

Command 4

```
db.Students.find().pretty()
```

Output

```
Displays the 3 inserted students.
```

PRACTICAL 22

Question: Find students whose CGPA is greater than 8.5.

Command 1

```
db.Students.find(
  { CGPA: { $gt: 8.5 } },
  { _id: 0 }
)
```

Output

```
Tulika— CGPA 9
```

PRACTICAL 23

Question: Find students whose CGPA is less than 9.

Command 1

```
db.Students.find(
  { CGPA: { $lt: 9 } },
  { _id: 0 }
)
```

Output

```
Tulika – CGPA 8.5
Shikha – CGPA 8.2
```

PRACTICAL 24

Question: Find students whose CGPA is greater than or equal to 8.5.

Command 1

```
db.Students.find(
  { CGPA: { $gte: 8.5 } },
  { _id: 0 }
)
```

Output

```
Tulika – CGPA 8.5
Harsh Sharma – CGPA 9
```

PRACTICAL 25

Question: Find BCA students whose CGPA is greater than or equal to 8.5.

Command 1

```
db.Students.find(
  {
    Department: "BCA",
    CGPA: { $gte: 8.5 }
  },
  { _id: 0 }
)
```

Output

```
Tulika – BCA – CGPA 8.5
Harsh Sharma – BCA – CGPA 9
```